Assignment 3A: Global Baseline Estimate LLM Transcription

2026-02-15

LLM Transcript and Citation

Tool Used: ChatGPT (OpenAI, GPT-5.2)
Purpose: Implementing a Global Baseline Estimate recommender system in R using movie ratings stored in PostgreSQL
Course: DATA 607 – Assignment 3A (Global Baseline Estimate)
Date Accessed: February 15 2026

Citation:
OpenAI. (2026). *ChatGPT (GPT-5.2)* [Large language model]. https://chat.openai.com.

I used ChatGPT to help implement the Global Baseline Estimate recommendation system. Specifically, I used it to (1) generate predicted ratings using the baseline formula, (2) produce movie recommendations for unseen items, and (3) create visualizations to interpret model behavior.

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[1] Conversation with ChatGPT (Generating predicted ratings)

How do I compute predicted ratings using the Global Baseline Estimate formula?

Assistant (ChatGPT):
You can combine the global mean, user bias, and movie bias using:

Suggested code:

```
predicted_observed <- ratings_clean %>%
  left_join(user_bias, by = "user_id") %>%
  left_join(movie_bias, by = "movie_id") %>%
  mutate(predicted_rating = mu + b_u + b_i)
```

[2] Conversation with ChatGPT (Generating recommendations for unseen movies)

How can I generate recommendations for movies users have not rated?

Assistant (ChatGPT): Create all user–movie combinations, remove already-rated movies, compute predicted ratings, and select the top-N movies per user.

Suggested code:

```
recommendations_all <- all_scored %>%
  filter(is.na(rating)) %>%
  group_by(user_id, name) %>%
  arrange(desc(predicted_rating), .by_group = TRUE) %>%
  slice_head(n = 5)
```

[3] Conversation with ChatGPT (Visualization for interpretation)

Any visualization ideas to make this more interesting?

Assistant (ChatGPT): You can visualize model components such as movie bia, user bias or compare actual and predicted ratings.

Suggested Code:

```
movie_bias %>%
  left_join(movies_df, by = "movie_id") %>%
  ggplot(aes(x = reorder(title, b_i), y = b_i)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Movie Bias (b_i) from Global Baseline Estimate",
    x = "Movie",
    y = "Movie Bias"
  )

user_bias %>%
```

```r
  left_join(users_df, by = "user_id") %>%
  ggplot(aes(x = reorder(name, b_u), y = b_u)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "User Bias (b_u)",
    x = "User",
    y = "User Bias"
  )

ggplot(predicted_observed, aes(x = rating, y = predicted_rating)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(
    title = "Actual vs Predicted Ratings",
    x = "Actual Rating",
    y = "Predicted Rating"
  )
```

ChatGPT is an AI and can mistakes